



ESCUELA DE INGENIERÍA
FACULTAD DE INGENIERÍA

Lecture 21: Monopoly II

Patricia Galilea A.

ICS2523 Microeconomics

Lecture 21

Last lecture, we

- Started Monopoly
 - Definition
 - Barriers to entry
 - Monopoly's profits
 - Resource allocation problem

In today's lecture, we will continue studying **Monopoly**:

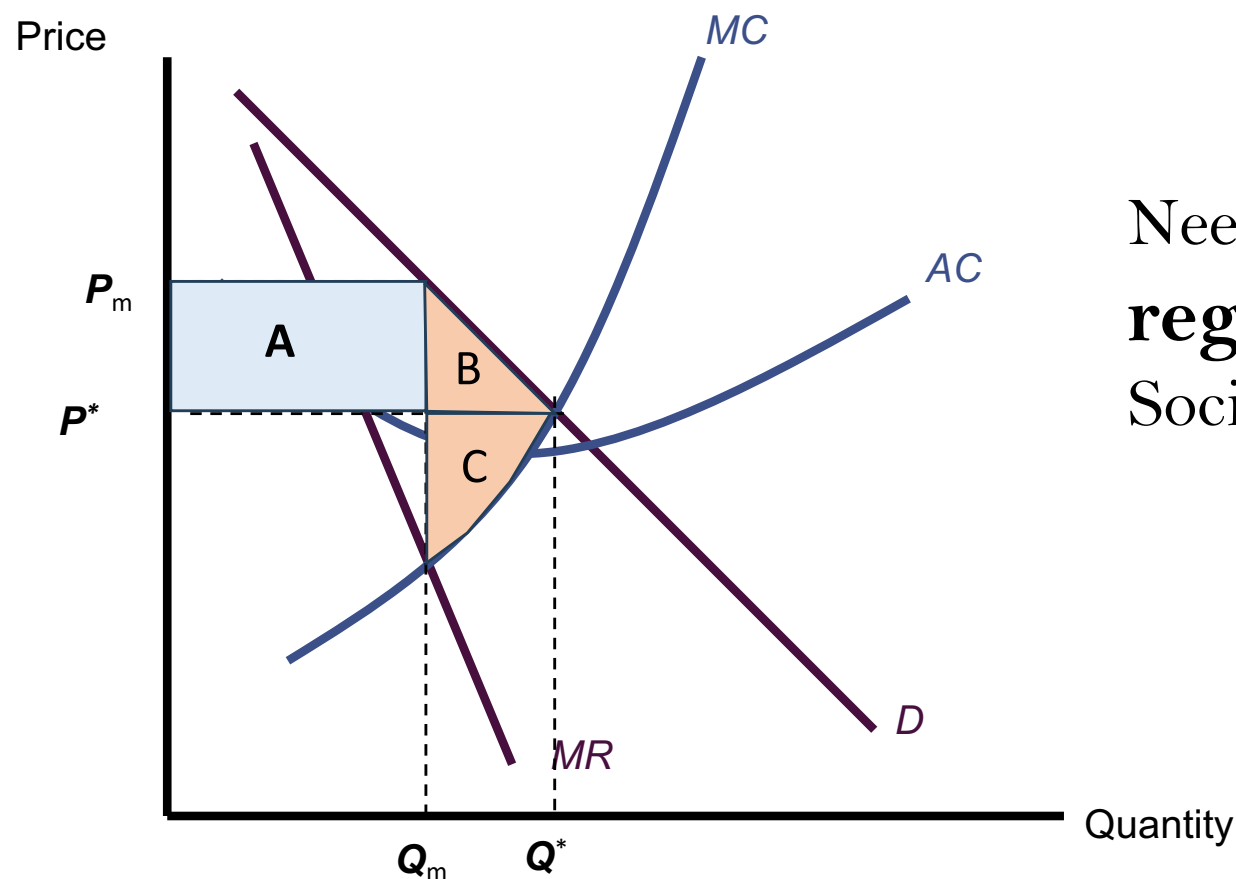
- Regulation
- Price discrimination

Monopoly: Regulation

Market Failures

5. Regulation of Monopoly

When a monopoly operates a market, market price no longer conveys accurate information about production costs $\rightarrow$ resources will be misallocated $\rightarrow$ deadweight loss (loss in efficiency)



Need for **government regulation** to improve Social Welfare

5. Regulation of Monopoly



BUSCADOR CONTACTO ESP ENG

≡ MENÚ



ANTIMONOPOLIOS

[Home](#) / [Antimonopolios](#) / [Qué Hacemos](#)

[Qué Hacemos](#)

[Guías](#)

[Etapas de una Investigación](#)

[Publicaciones](#)

[Interlocking](#)

[Participación en Competidores](#)

Qué Hacemos

La División Antimonopolios (DAM) es una división persecutoria de la FNE que investiga conductas contrarias al Decreto Ley N° 211 (DL 211), excepto las correspondientes a la División Anti-Carteles, y recomienda al Fiscal Nacional Económico cursos de acción con el objeto de promover y defender la libre competencia en los mercados.

Las principales conductas investigadas por la DAM son los abusos de posición dominante, contemplados en la letra b) y c) del artículo 3 del DL 211.

Asimismo, a la División Antimonopolios le corresponde investigar conductas predatorias, de competencia desleal, interlocking, así como otras que podrían vulnerar el inciso primero del artículo 3 del DL 211, dentro las que destacan las restricciones verticales y los acuerdos de cooperación entre competidores.

Por otro lado, en esta División se desarrollan fiscalizaciones tendientes a velar por el cumplimiento de los fallos y decisiones del TDLC, o los tribunales de justicia, en las materias a que se refiere el DL 211.

Finalmente, la DAM puede también actuar con ocasión de disposiciones sobre competencia existentes en normas regulatorias sectoriales, tales como: Ley N° 19.542, que moderniza el Sector Portuario Estatal; Ley N° 19.733, sobre Libertades de Opinión e Información y Ejercicio del Periodismo (Ley de Prensa); DFL N° 1.122 de 1981, Código de Aguas; DFL N° 70 de 1988, Tarifas Servicios Sanitarios ; DFL N°4, Ley General de Servicios Eléctricos; DFL N° 323, Ley de Servicios de Gas; Ley N° 18.168, Ley General de Telecomunicaciones, entre otras.

<https://www.fne.gob.cl/antimonopolios/>

5. Regulation of Monopoly

Regulation of natural monopolies is an important subject in applied economic analysis → utilities, communications, and transportation industries are highly regulated

<http://www.youtube.com/watch?v=lZfbZDKOhLw&feature=related>

→ Importance of **devising regulatory procedures** that induce these industries to operate in a desirable way

1) Marginal cost pricing: prices should reflect marginal costs of production

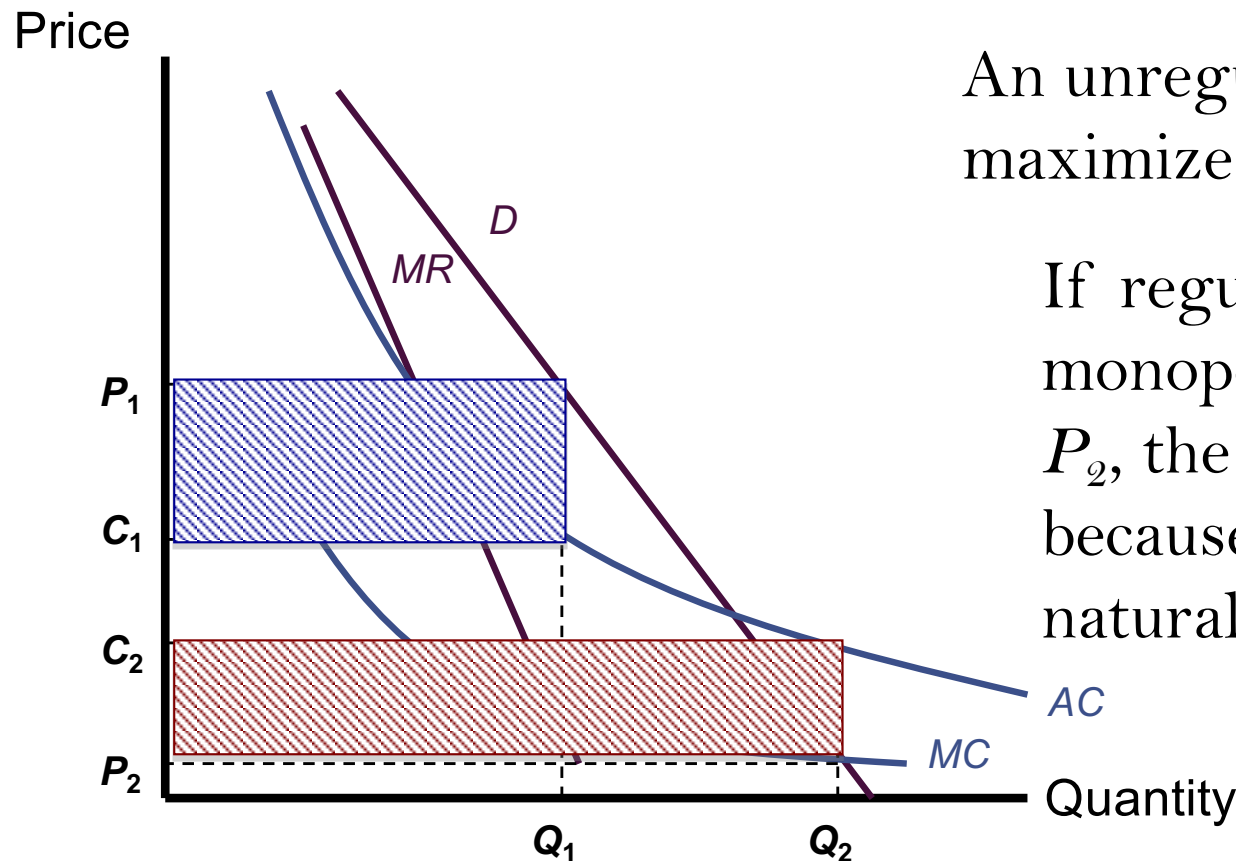
Problem?

This will require natural monopolies to operate at loss, as the price would be below the average cost curve

Dilemma: abandon goal of marginal cost pricing (and let $P=AC$) or government must subsidize monopoly

5. Regulation of Monopoly

1) **Marginal cost pricing:** prices should reflect marginal costs of production



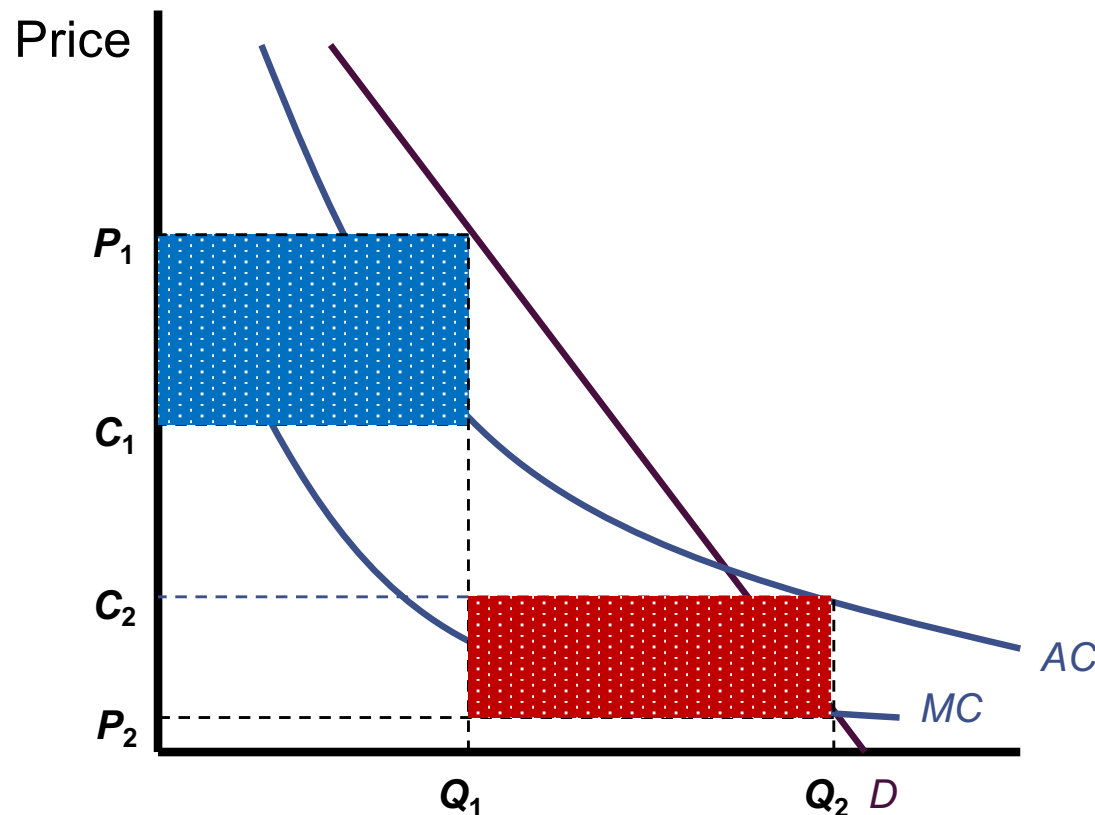
An unregulated monopoly will maximize profit at Q_1 and P_1

If regulators force the monopoly to charge a price of P_2 , the firm will suffer a loss because $P_2 < C_2$ if it is a natural monopoly ($MC < AC$)

5. Regulation of Monopoly

2) Two-tier pricing systems: implementation of a multi-price system, monopolist is permitted to charge some users a high price while maintaining a low price for marginal users

→ Consumers paying the high price subsidize the losses of the low-price consumers



Graphs by South-Western,
a division of Thomson
Learning. All rights
reserved.

5. Regulation of Monopoly

3) Rate of return regulation:

Permits the monopoly to charge a price above MC that is sufficient to earn a “fair” rate of return on investment (in capital)

What is a “fair” rate of return? How can it be measured?

Besides, a firm’s capital stock is difficult to value

Difficulty in agreeing on a set of numbers to be used in rate of return calculations leads to delays in regulatory response to changes in cost and other markets conditions

5. Regulation of Monopoly

3) Rate of return regulation:

Who wins?

- Lawyers, accountants, and economic consultants, because of the regulatory lag (delays of a year or more).
- Sometimes this regulatory lag worked to the advantage of regulatory firms-electric utilities (the 1950s and 1960s when costs were falling, usually because of economies of scale)
- But in the 1970s, the situation changed, and regulatory lag worked to the detriment of regulated firms (oil prices rose sharply)
- Since 1990 regulatory environment has changed in the USA, and telecommunications has been deregulated as electric utilities.

Monopoly: Price Discrimination

Market Failures

6. Monopoly: Price Discrimination

In some cases, a monopoly may increase profits by departing from a single-price policy for its output.

Price discrimination → is the possibility of selling **identical goods** at different prices (taking advantage of the willingness to pay)

Examples?

There are different types of price discrimination:

- First-degree price discrimination
- Third-degree price discrimination through market separation
- Second-degree price discrimination through price schedules
- Second-degree price discrimination through quality differentiation

6. Monopoly: Price Discrimination

Is it always feasible to apply a price discrimination policy?

- Depends on the inability of buyers of the good to practice arbitrage
- If there is an absence of transactions or information costs, the “law of one price” implies that a homogeneous good must sell everywhere for the same price
 - Price discrimination schemes are doomed to failure because demanders who can buy (from the monopoly) at lower prices will be the more attractive sources of the good for those who pay high prices than the monopoly
- When **resale** is costly or can be prevented entirely, price discrimination becomes possible

6. Monopoly: Price Discrimination

Which firms can price discriminate?

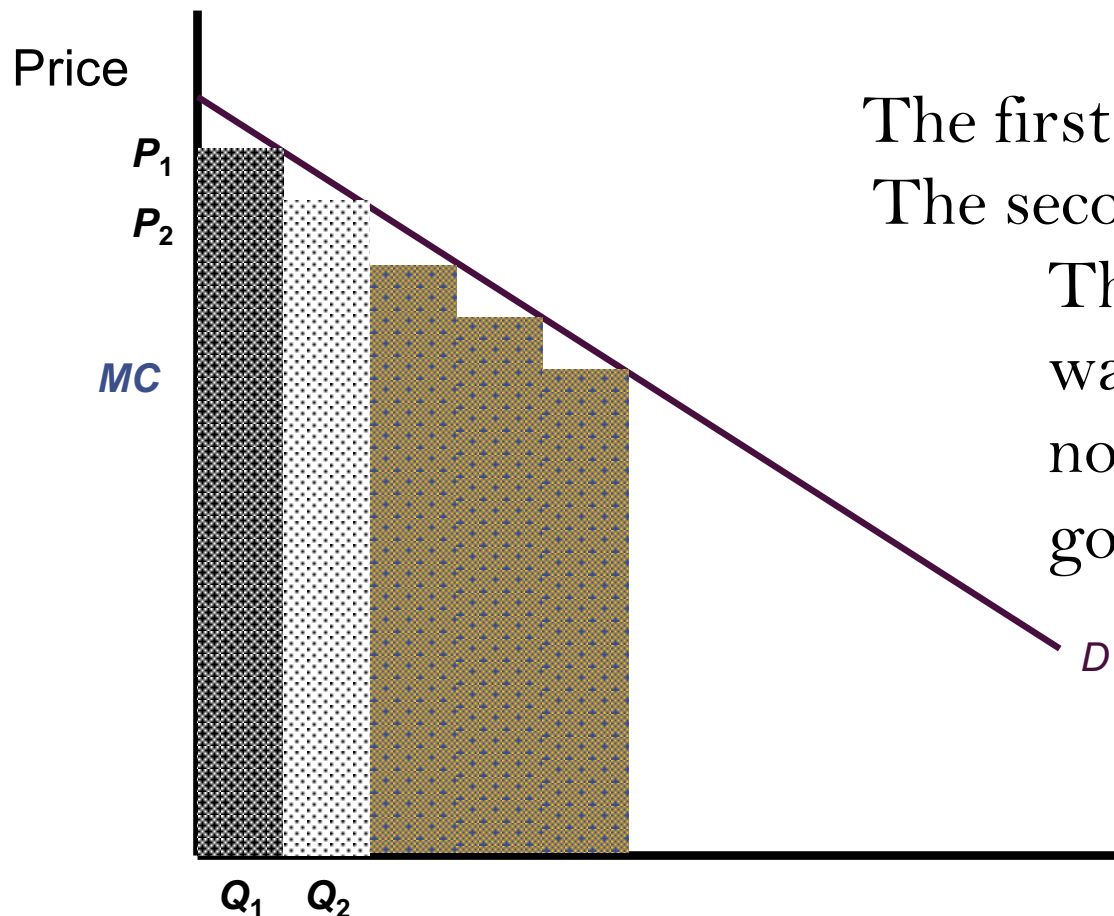
Three conditions must be met:

1. A firm must have market power
2. For a firm to profitably discriminate, groups of consumers or individual consumers must have demand curves that differ, and the firm must be able to identify how its consumers' demand curves differ
3. A firm must be able to prevent or limit resale

6. Monopoly: Price Discrimination

First-degree of Perfect Price Discrimination

If the monopolist can identify each buyer, then it may be possible to charge each the maximum price they are willing to pay for the good → through first-degree price discrimination. The monopolist would extract all available consumer surplus.



The first buyer pays P_1 for Q_1 units

The second buyer pays P_2 for $Q_2 - Q_1$ units

The monopolist will continue this way until the marginal buyer is no longer willing to pay the good's marginal cost

Activity 1

In a market with a monopolist applying First-Degree price discrimination, how much is the deadweight loss (compared with a perfect competition situation)?

- a. Equal to the consumer surplus
- b. Equal to the producer surplus
- c. Equal to the sum of the consumer and producer surplus
- d. Equal to zero



Código de evento Wooclap
CNKGJ

<https://app.wooclap.com/CNKGJZ?from=instruction-slide>

6. Monopoly: Price Discrimination

What happens when consumers are heterogeneous?

In first-degree price discrimination, monopolists must know the demand function for each potential buyer.

If a monopolist **cannot separate buyers** into several identifiable markets (examples?)

→ **Second-degree price discrimination**

If a monopolist **can separate buyers** into several identifiable markets (examples?)

→ **Third-degree price discrimination**

Pursues a separate monopoly pricing policy in each market.

What Is Pink Tax, Is It Unfair For Women In India?

While the income gap and pension gap already puts women at a disadvantage, now the pink tax further contributes to economic inequality between men and women. This issue needs to be scrutinized by the government and addressed at the earliest.



— Pink Tax is commonly used to describe a situation wherein things associated with women are priced higher than those with men. Generally speaking, women pay more than males for similar items. The pink tax is not enforced by the government but by the businesses selling the products. According to experts, the pricing disparity is due to market forces.

According to a study by the New York City Department of Consumer Affairs in 2015, women's products are typically more expensive than men's without a proper reason. The study concluded that products explicitly marketed for women cost seven per cent more on average than products sold for men. This discrepancy includes apparel, toys, and healthcare products, among other things. The largest discrepancy came to personal care/hygiene products, where women's products cost 13 per cent more than men's.

<https://www.outlookindia.com/business/what-is-pink-tax-is-it-unfair-for-women-in-india--news-268718>

Third degree price discrimination

example: **Pink tax**

Several countries have investigated the pink tax, including Argentina, France, Germany, the UK, Australia, and Italy. For example, in the UK, it was found that women and girls were being charged, on average, 37 per cent more for toys, cosmetics, and clothes than their male counterparts. In the UK, girls' school uniforms were also 12 per cent more expensive than boys' uniforms. In Singapore, a check by the Sunday Times on 10 companies found that women pay more for some products and services, such as dry cleaning and razors, offered by around half of these companies. Additionally, women in Singapore have to pay more premiums for Careshield Life, a national long-term care insurance scheme introduced by the government.

<https://www.outlookindia.com/business/what-is-pink-tax-is-it-unfair-for-women-in-india--news-268718>



<https://www.huffpost.com/entry/pink-tax-examples> | 5d24da77e4b0583e482850f0

6. Monopoly: Price Discrimination

Third-degree price discrimination through market separation

In first-degree price discrimination, monopolist must know the demand function for each potential buyer

In the third-degree price discrimination, we will assume that the monopoly **can separate buyers** into several identifiable markets (examples?) and pursue a separate monopoly pricing policy in each market

→ only needs price elasticities of demand

→ Monopoly sets a price in each market according to the inverse elasticity rule

$$MR = P + \frac{\partial P}{\partial Q} Q = MC$$

$$MR = P + P \left(\frac{1}{\epsilon_d} \right) = MC$$

6. Monopoly: Price Discrimination

Third-degree price discrimination through market separation

$$MR = P + P \left(\frac{1}{\varepsilon_d} \right) = MC$$

Suppose marginal cost is the same in all markets, what is the pricing policy if there is 2 types of consumers (i and j)?

$$P_i \left(1 + \frac{1}{\varepsilon_i} \right) = P_j \left(1 + \frac{1}{\varepsilon_j} \right)$$

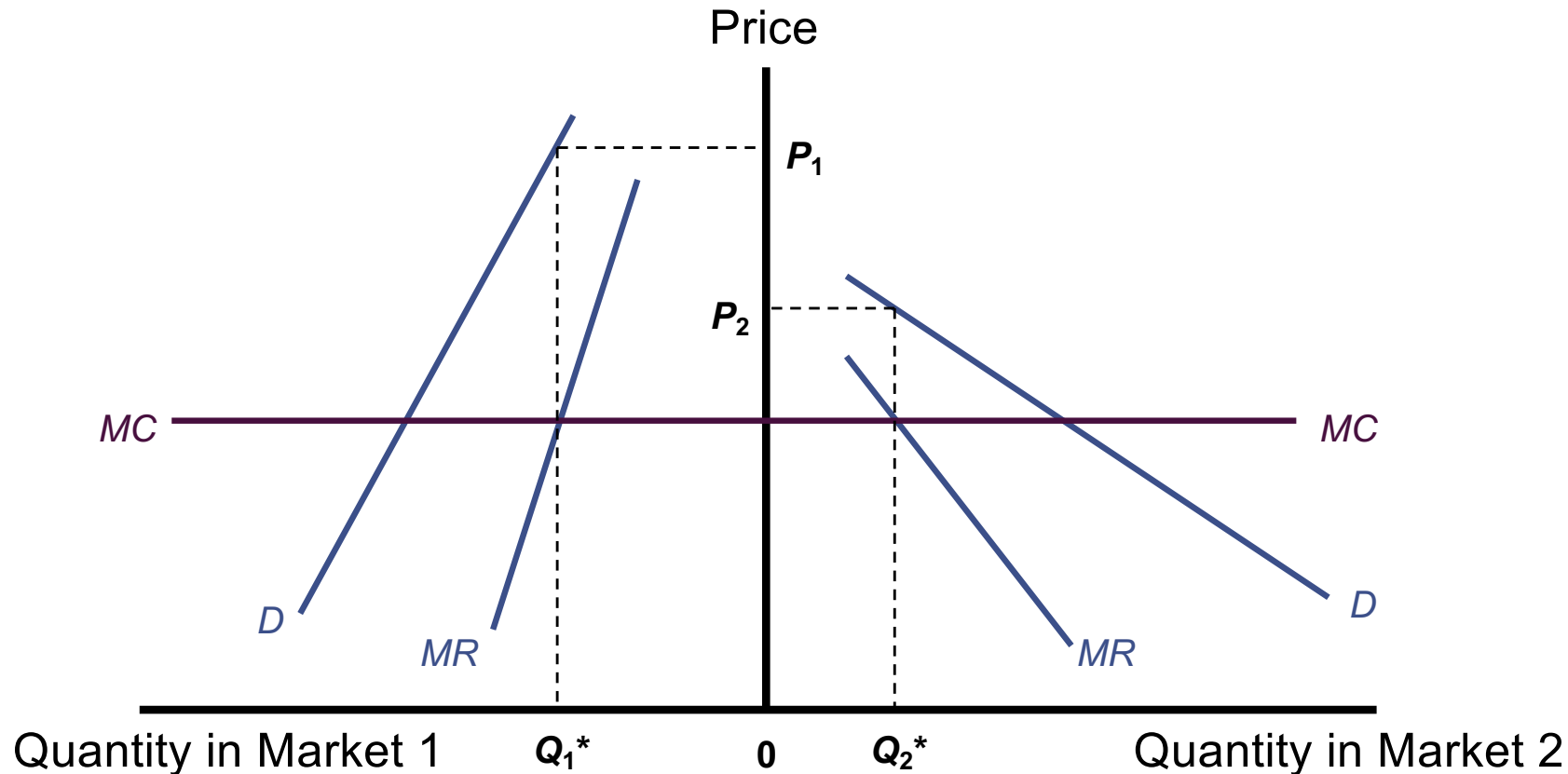
$$\frac{P_i}{P_j} = \frac{\left(1 + \frac{1}{\varepsilon_j} \right)}{\left(1 + \frac{1}{\varepsilon_i} \right)}$$

In which market price will be higher?

6. Monopoly: Price Discrimination

Third-degree price discrimination through market separation

In which markets price will be higher?



6. Monopoly: Price Discrimination

Third-degree price discrimination through market separation

What happens to welfare under third-degree price discrimination?

6. Monopoly: Price Discrimination

Second-degree price discrimination through price schedules

What happens if the monopolist cannot separate buyers into categories?

He knows that **there are different consumers**, but he cannot distinguish between them

Alternative approach: monopoly will choose **a price schedule** that provides incentives for demanders to separate themselves **depending on how much they wish to buy**:

- Quantity discounts
- Minimum purchase requirements (cover charges)
- Examples?

6. Monopoly: Price Discrimination

Second-degree price discrimination through price schedules

Example: Two-part tariff

Consumers must **pay a fixed fee** for the right to consume a good and a **uniform price for each unit consumed**.

The prototype case (first studied by Walter Oi) is an amusement park that sets a basic entry fee coupled with a stated marginal price for each amusement used.

So, each demander must pay a tariff T to purchase q units of a good:

$$T(q) = a + pq$$

Where a is the fixed cost and p is the marginal price to be paid.

6. Monopoly: Price Discrimination

Second-degree price discrimination through price schedules

Example: Two-part tariff

Thus, the average price paid by any demander is:

$$\bar{p} = \frac{T}{q} = \frac{a}{q} + p$$

This tariff is only feasible when those who pay low average prices (those for whom q is large) cannot resell the good to those who must pay high average prices (those for whom q is small)

6. Monopoly: Price Discrimination

Second-degree price discrimination through quality differentiation

Alternative approach: monopoly offer different a product with different **qualities** that provides incentives for demanders to separate themselves depending on what they want to buy:

- Airlines: first, business, and economic class
- Regular and professional versions of software

Homework Lecture 21

Suppose that a monopoly producer of widgets has a constant marginal cost of $c = 6$ and sells its products in two separated markets whose inverse demand functions are:

$$P_1 = 24 - Q_1 \text{ and } P_2 = 12 - \frac{Q_2}{2}$$

If the monopolist uses Third-degree price discrimination,

- a) Which are the profit-maximizing price-quantity combinations in these two markets?
- b) Find the profits
- c) Calculate the total deadweight loss (compared with the competitive output)
- d) If the monopolist is constrained by the government to charge a single price, what will happen with welfare?

Individual Activity

On a piece of paper, write down **2-3 concepts** that you learnt today. Please write down the concept and what you learnt about it in your own words.

If you have any **questions** about what we discussed today, write them down too. We Will go through them at the beginning of the next class.

Finally, if there is a word you didn't understand, write it down so it can be part of the **glossary**.